

SHAMARY, Russian Federation

WMO: 283340

Lat: 57.350N

Lon: 58.217E

Elev: 244

StdP: 98.43

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.7	-29.4	-35.8	0.1	-32.5	-32.4	0.2	-29.3	8.2	-8.6	6.5	-10.3	0.4	90	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	29.4	19.8	27.5	18.8	25.6	17.8	20.8	27.2	19.9	26.0	18.9	24.2	2.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.9	23.9	17.7	13.0	22.4	16.7	12.2	21.6	61.3	27.3	57.9	26.1	54.5	24.4	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.7	4.7	DB	-37.0	31.9	4.0	1.9	-40.0	33.3	-42.3	34.4	-44.6	35.5	-47.5	36.9	(4)
(5)			WB	-37.1	22.4	4.0	1.6	-40.0	23.6	-42.4	24.6	-44.7	25.5	-47.6	26.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.1	-14.4	-12.7	-4.8	3.1	10.6	15.5	17.7	15.2	9.6	2.6	-6.1	-12.3	(6)	
	DBStd	12.70	7.78	7.39	5.12	5.10	4.88	4.41	4.04	4.15	4.41	4.58	7.14	8.14	(7)	
	HDD10.0	3597	757	636	459	213	53	7	1	4	59	233	483	691	(8)	
	HDD18.3	6020	1015	869	718	458	243	106	63	116	263	488	733	949	(9)	
	CDD10.0	706	0	0	0	6	72	173	239	166	48	3	0	0	(10)	
	CDD18.3	86	0	0	0	0	3	21	42	18	1	0	0	0	(11)	
	CDH23.3	979	0	0	0	1	64	251	455	193	16	0	0	0	(12)	
	CDH26.7	244	0	0	0	0	4	61	124	54	1	0	0	0	(13)	
(14) Wind	WSAvg	1.8	1.8	1.9	2.1	2.0	1.9	1.7	1.3	1.3	1.5	2.0	1.9	1.9	(14)	
(15) Precipitation	PrecAvg	734	52	40	42	40	51	81	83	83	67	73	64	59	(15)	
	PrecMax	1006	106	100	86	94	108	174	186	183	129	121	106	127	(16)	
	PrecMin	600	4	14	3	0	16	17	23	19	20	15	23	27	(17)	
	PrecStd	97	22	20	21	25	23	41	44	38	31	30	24	25	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.1	2.2	9.7	21.8	27.3	31.2	31.7	31.1	25.6	17.4	8.4	1.3	(19)	
		MCWB	0.1	0.2	4.4	12.1	15.4	19.8	20.6	20.2	16.2	11.3	6.7	0.5	(20)	
	2%	DB	-0.5	0.5	6.0	17.3	25.3	28.3	29.6	27.9	22.1	14.2	5.4	0.4	(21)	
		MCWB	-1.2	-0.6	1.9	9.9	14.9	18.7	20.3	19.5	15.3	9.6	4.3	-0.2	(22)	
	5%	DB	-1.9	-0.7	3.6	14.2	22.8	26.3	28.0	25.1	19.0	11.5	3.1	-0.5	(23)	
		MCWB	-2.5	-1.7	0.8	7.9	13.8	17.8	19.6	18.3	14.2	8.3	2.1	-1.2	(24)	
	10%	DB	-3.8	-2.5	1.9	11.3	20.3	24.0	26.0	22.6	16.5	9.4	1.5	-2.0	(25)	
		MCWB	-4.5	-3.5	0.0	6.0	12.6	16.9	18.9	17.3	12.9	7.2	0.7	-2.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.4	0.6	5.0	12.8	17.3	21.2	22.7	21.2	17.8	11.9	6.6	1.0	(27)	
		MCDB	1.0	1.8	9.1	20.4	24.6	28.1	29.4	28.1	22.6	15.1	7.9	1.1	(28)	
	2%	WB	-1.2	-0.4	2.6	10.4	15.7	19.7	21.4	20.2	15.8	10.1	4.3	-0.1	(29)	
		MCDB	-0.7	0.5	5.0	16.5	22.7	26.3	27.4	26.4	20.3	13.2	5.2	0.3	(30)	
	5%	WB	-2.5	-1.7	1.3	8.5	14.5	18.7	20.4	19.0	14.5	8.6	2.3	-1.3	(31)	
		MCDB	-1.9	-0.8	3.0	13.5	21.2	24.5	26.5	24.0	18.2	11.0	3.0	-0.5	(32)	
	10%	WB	-4.5	-3.4	0.1	6.7	13.4	17.6	19.4	17.7	13.2	7.2	0.8	-2.8	(33)	
		MCDB	-3.9	-2.5	1.8	10.5	19.1	23.0	25.1	21.7	16.1	9.3	1.4	-2.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	9.0	9.2	10.2	11.8	10.8	10.8	9.6	8.4	5.5	5.2	6.4	(35)	
		MCDBR	5.5	6.4	11.1	15.3	16.8	14.5	14.5	13.8	12.8	11.0	5.4	4.6	(36)	
	5% WB	MCWBR	5.3	5.6	7.9	8.9	7.8	6.6	6.1	6.4	6.9	7.0	4.6	4.4	(37)	
		MCWBR	5.2	5.9	8.4	13.7	14.1	12.8	13.0	12.8	11.4	9.7	5.0	4.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.288	0.314	0.370	0.406	0.400	0.395	0.421	0.434	0.379	0.349	0.307	0.284	(40)	
		taud	2.006	2.014	1.927	2.032	2.264	2.338	2.265	2.200	2.324	2.262	2.093	1.968	(41)	
	Ebn at Noon		529	694	747	794	835	845	810	764	757	664	506	406	(42)	
		Edh at Noon	61	95	136	144	123	116	122	122	93	75	56	48	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.58	1.48	2.75	4.03	5.01	5.34	5.32	3.99	2.42	1.07	0.49	0.33	(44)	
	RadStd		0.09	0.22	0.28	0.51	0.54	0.50	0.55	0.45	0.22	0.19	0.09	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page